

PRE-TEST

Multi-Agent System Design

Agentic AI Coding Fitness | Week 9

Name: _____ Date: _____

Score: _____ / 50 Time: _____ minutes

Instructions

Total questions	50 multiple-choice questions, one correct answer each
Scoring	1 point per correct answer. No penalty for wrong answers. Maximum score: 50 points.
Time limit	45 minutes recommended (approximately 54 seconds per question)
Format	Circle or tick the letter (A, B, C, or D) for the answer you believe is correct.
Topics covered	MAS Fundamentals 8 Design Questions FIPA-ACL Agent Types & ADK Collaboration & Environment A2A Protocol
Purpose	This pre-test benchmarks your current knowledge. Results help identify which workshop sections to focus on. There are no "pass" or "fail" grades.
After the workshop	The same 50 questions may be used as a post-test to measure your learning gain.

Question Categories

Category	Questions	Topics
MAS Fundamentals	Q1 - Q10	What is MAS, emergence, autonomy, when to use MAS
MAS Design Framework	Q11 - Q20	The 8 design questions, agent sizing, observability
Agent Types & ADK Patterns	Q21 - Q29	SequentialAgent, ParallelAgent, LoopAgent, AgentTool, HITL
FIPA-ACL Communication	Q30 - Q38	Performatives, Contract Net, Request-Reply, ontology
Collaboration & Environment	Q39 - Q44	Cooperative, competitive, observable, blackboard, state estimation
A2A Protocol & Integration	Q45 - Q50	A2A vs MCP, Agent Card, cross-framework, session.state

1	MAS Fundamentals	What is the most accurate definition of a Multi-Agent System (MAS)?
	A	A single AI model that processes multiple input prompts in parallel
	B	A system of multiple interacting autonomous agents, each with its own goals, perceptions, and capabilities
	C	A distributed database where multiple servers handle different tables
	D	A neural network with multiple hidden layers and attention heads
2	MAS Fundamentals	Which of the following is NOT a benefit of using a Multi-Agent System compared to a single agent?
	A	Modularity — each agent handles a focused concern
	B	Specialization — agents can have domain-specific tools and knowledge
	C	Lower LLM token cost — fewer total API calls needed
	D	Fault tolerance — if one agent fails, others continue working
3	MAS Fundamentals	According to Wooldridge's agent theory, which property describes an agent that acts on behalf of users without constant human direction?
	A	Reactivity — responding to environmental changes
	B	Proactivity — taking initiative to achieve goals
	C	Social ability — communicating with other agents
	D	Autonomy — making decisions independently
4	MAS Fundamentals	In MAS theory, "emergence" refers to:
	A	A new agent type that replaces all other specialized agents
	B	Complex system-level behavior arising from interactions between agents, not explicitly programmed into any single agent
	C	The process of an LLM generating new agent code automatically
	D	An error state when two agents simultaneously request the same resource
5	MAS Fundamentals	Which real-world system is the BEST analogy for a cooperative Multi-Agent System?
	A	A single surgeon performing an operation alone
	B	A calculator executing one formula at a time
	C	A hospital where doctors, nurses, pharmacists, and specialists collaborate on patient care
	D	A spreadsheet with multiple formulas in separate cells
6	MAS Fundamentals	What is "coordination overhead" in a Multi-Agent System?
	A	The electricity cost of running multiple GPU servers
	B	The extra communication, synchronization, and conflict-resolution work required between agents
	C	The number of lines of code needed to define each agent
	D	The delay introduced by rate limiting on external APIs

7	MAS Fundamentals	When should you prefer a single agent over a Multi-Agent System?
A	When the task requires real-time parallel data collection from multiple sources	
B	When the task needs specialized expertise across three distinct domains	
C	When a simple sequential task comfortably fits within one agent's context and capabilities	
D	When the system must continue working if one component fails	
8	MAS Fundamentals	An "agent" in MAS is best described as an entity that:
A	Executes one fixed function and returns a result	
B	Perceives its environment, makes decisions, and takes actions to achieve its goals	
C	Stores data in a structured format for other systems to query	
D	Generates images or text from user prompts	
9	MAS Fundamentals	What is the minimum number of interacting entities needed to technically form a "Multi-Agent System"?
A	1	
B	2	
C	5	
D	10	
10	MAS Fundamentals	Which scenario BEST demonstrates why MAS is needed over a single agent?
A	A chatbot answering simple FAQ questions from a knowledge base	
B	A system that simultaneously monitors energy in 50 hotel zones, each requiring different control logic and local sensor access	
C	A tool that converts PDF documents to text format	
D	An AI that generates captions for social media images	
11	MAS Design Framework	Before designing a MAS, the FIRST question you should answer is:
A	Which cloud provider to deploy on	
B	Which programming language and LLM framework to use	
C	What specific problem you want to solve and what measurable success looks like	
D	How many API tokens your budget allows per month	
12	MAS Design Framework	You need to simultaneously fetch weather data, stock prices, and news headlines. Which reason BEST justifies using a MAS?
A	It is cheaper to run three small agents than one large agent	
B	Single agents cannot make HTTP requests	
C	Multiple independent parallel workstreams benefit from concurrent execution	
D	The data sources require three different programming languages	

13	MAS Design Framework	When defining each agent (Design Question 4), which four properties are most important to specify?
A	Name, cloud region, deployment date, monthly cost	
B	Goal, Backstory/Persona, Tools it can call, Output it produces	
C	Framework, LLM model, temperature setting, max tokens	
D	Color code, icon, font size, display position in UI	
14	MAS Design Framework	A fraud detection agent only sees transaction patterns but cannot observe the customer's actual intent or location. This environment is:
A	Fully observable — the agent sees all transaction data	
B	Partially observable — key state variables (intent, location) are hidden	
C	Static — the environment does not change during analysis	
D	Deterministic — the same transaction always produces the same risk score	
15	MAS Design Framework	When is "state estimation" needed in a MAS design?
A	When agents have too many tools and need to choose between them	
B	When the environment is partially observable and agents must infer hidden state from indirect signals	
C	When the LLM model does not know the current date	
D	When two agents produce conflicting outputs	
16	MAS Design Framework	Design Question 6 asks about agent collaboration mode. In a market-based MAS, multiple delivery agents submit cost bids for a logistics contract. This demonstrates which mode?
A	Cooperative — all delivery agents share the same goal	
B	Competitive — agents bid against each other for the contract	
C	Adversarial — agents actively try to sabotage each other	
D	Hierarchical — a central agent assigns tasks without bidding	
17	MAS Design Framework	According to the design principle, which question should you ask yourself after listing all your planned agents?
A	Which agent should have the highest LLM temperature?	
B	Could any two agents be merged without violating separation of concerns?	
C	Which agent should run first alphabetically?	
D	How many tokens will each agent use per day?	
18	MAS Design Framework	Design Question 8 asks how MAS fits your architecture. What should a production MAS expose to a frontend application?
A	Direct access to agent Python objects via import statements	
B	Raw LLM prompt/response logs	
C	A well-defined API surface such as REST, WebSocket, or event stream	
D	Database connection strings for each agent's memory store	

19	MAS Design Framework	A developer says "I'll build a MAS with 15 agents because each sensor type needs its own agent." What is the most important concern to raise?
A	Whether 15 is an even number	
B	Whether the cloud budget supports 15 agents	
C	Whether 15 agents can be merged into fewer without losing necessary specialization — more agents means more coordination overhead	
D	Whether all 15 agents should use the same LLM model	
20	MAS Design Framework	You are designing a food safety monitoring MAS. The system must both continuously monitor temperatures AND generate regulatory reports. What does this suggest about agent design?
A	One MonitoringAgent handles both — less coordination overhead	
B	Two separate agents: TempMonitorAgent (continuous loop) and ReportAgent (triggered on event) — different execution patterns require different agent types	
C	Monitoring should be done manually; only reporting needs an agent	
D	Both tasks should be combined into a single LLM prompt with two instructions	
21	Agent Types & ADK Patterns	In Google ADK, which agent type executes sub-agents one after another in a fixed sequence, passing shared state between them?
A	ParallelAgent	
B	LoopAgent	
C	SequentialAgent	
D	CoordinatorAgent	
22	Agent Types & ADK Patterns	Which ADK agent type is best for simultaneously fetching weather data from three different APIs to minimize total latency?
A	SequentialAgent — runs all fetches in order	
B	LoopAgent — repeats until all fetches succeed	
C	ParallelAgent — executes all sub-agents concurrently	
D	LlmAgent — uses LLM to decide which API to call	
23	Agent Types & ADK Patterns	A LoopAgent in Google ADK stops executing when:
A	The first sub-agent raises a Python exception	
B	All agents produce identical outputs	
C	max_iterations is reached OR any sub-agent returns an event with escalate=True	
D	The LLM model rate limit is exceeded	

24	Agent Types & ADK Patterns	In Google ADK, how does an LLM Agent dynamically route a task to a different specialist agent?
A By writing the agent name to a shared file and polling		
B By calling <code>transfer_to_agent(agent_name="TargetAgent")</code>		
C By raising a Python <code>RuntimeError</code> with the target agent name		
D By restarting the entire workflow from the root agent		
25	Agent Types & ADK Patterns	What is the purpose of the <code>output_key</code> parameter on a Google ADK <code>LlmAgent</code> ?
A To encrypt the agent's response before storing it		
B To automatically save the agent's final response to a named key in <code>shared.session.state</code>		
C To specify which API key the agent should use for LLM calls		
D To define the priority order when multiple agents compete for resources		
26	Agent Types & ADK Patterns	Which ADK mechanism allows a parent LLM agent to call a child agent as an explicitly invocable tool?
A <code>LoopAgent</code> with shared memory		
B <code>AgentTool</code> — wraps a <code>BaseAgent</code> as a callable tool in the parent's tool list		
C <code>SessionManager</code> with agent binding		
D <code>DirectCall</code> — a built-in Python function for agent invocation		
27	Agent Types & ADK Patterns	A hospital MAS needs a doctor to approve medication orders over 1,000mg before an agent places them. Which ADK pattern handles this?
A <code>ParallelAgent</code> — the doctor and agent run simultaneously		
B <code>LoopAgent</code> — keep retrying until the doctor is available		
C Human-in-the-Loop via <code>FunctionTool</code> or <code>SecurityPlugin</code> with policy engine		
D <code>SequentialAgent</code> — the doctor is placed last in the sequence		
28	Agent Types & ADK Patterns	Which ADK pattern is BEST for improving output quality by having one agent write a draft and another critique it?
A <code>ParallelAgent</code> with two writers producing competing drafts		
B <code>LoopAgent</code> running the same agent multiple times until output is perfect		
C <code>SequentialAgent([WriterAgent, CriticAgent])</code> — the Generator-Critic pattern		
D Human-in-the-Loop where the human acts as the critic		
29	Agent Types & ADK Patterns	In Google ADK, how do agents within the same application share data between sequential steps?
A Via REST API calls between agent endpoints		
B Through a shared <code>session.state</code> dictionary accessible to all agents in the same invocation		
C By writing to a shared file on disk and reading from it		
D Via a message queue that agents subscribe to independently		

30	FIPA-ACL Communication	FIPA-ACL (Foundation for Intelligent Physical Agents - Agent Communication Language) defines:
A	A Python library for building AI agents	
B	Standard message types called "performatives" that agents use to communicate with defined meanings	
C	A cloud deployment specification for multi-agent systems	
D	An encryption protocol for securing agent-to-agent messages	
31	FIPA-ACL Communication	An EnergyBudgetAgent wants the HVACControlAgent to reduce zone temperature by 2 degrees. Which FIPA-ACL performative should it use?
A	INFORM — to tell HVAC the temperature has changed	
B	QUERY-IF — to ask if HVAC can reduce temperature	
C	REQUEST — to ask HVAC to perform the reduction action	
D	PROPOSE — to suggest the reduction as an option	
32	FIPA-ACL Communication	In the Contract Net Protocol, which performative does the initiator send to broadcast a task and invite bids from multiple agents?
A	REQUEST — directly ask each agent to do the task	
B	CALL-FOR-PROPOSAL (CFP) — broadcast the task for competitive bidding	
C	AGREE — signal that the task is assigned	
D	INFORM — announce what the task is	
33	FIPA-ACL Communication	A WarehouseAgent has successfully fulfilled an order. Which performative should it send to the OrderAgent?
A	FAILURE — to report the task outcome	
B	REFUSE — to confirm it has finished	
C	INFORM-DONE — to confirm the action was completed successfully	
D	AGREE — to accept the assignment again	
34	FIPA-ACL Communication	An InventoryAgent wants to know if a specific product is in stock. Which performative should it send to the StockAgent?
A	REQUEST — ask StockAgent to check inventory	
B	INFORM — tell StockAgent about the product	
C	QUERY-IF — ask whether a specific proposition (product in stock) is true	
D	PROPOSE — suggest what to do if out of stock	
35	FIPA-ACL Communication	An "ontology" in MAS communication design refers to:
A	The agent's training dataset and fine-tuning examples	
B	The shared vocabulary: objects, actions, attributes, and relations that all agents in the system understand	
C	The execution order in which agents are called	
D	The formal contract between the developer and the cloud provider	

36	FIPA-ACL Communication	What is the correct sequence for the basic FIPA-ACL "Request-Reply" protocol?
A PROPOSE -> ACCEPT -> CFP -> AGREE		
B REQUEST -> AGREE -> (action executes) -> INFORM-DONE or FAILURE		
C INFORM -> QUERY-IF -> PROPOSE -> REFUSE		
D CFP -> REQUEST -> FAILURE -> AGREE		
37	FIPA-ACL Communication	In hotel energy management ontology design, which best represents the "Actions" component of the ontology?
A Room, Zone, HVAC, Sensor, GuestProfile		
B temperature, setpoint, zone_id, occupancy_count		
C set_temperature(zone, value), read_occupancy(zone), override_schedule(zone, duration)		
D Zone has_sensor Sensor, Room belongs_to Zone		
38	FIPA-ACL Communication	A DeliveryAgent responds to a logistics CFP but cannot meet the deadline. Which performative should it send?
A FAILURE — reporting it cannot complete the delivery		
B PROPOSE — with a bid that includes a delay		
C REJECT-PROPOSAL — declining the initiator's terms		
D REFUSE — declining to make a proposal for the announced task		
39	Collaboration & Environment	In which collaboration mode do all agents work toward a single shared objective by dividing responsibilities among themselves?
A Adversarial — agents attempt to detect and counter each other		
B Competitive — agents self-interestedly bid for tasks		
C Cooperative — agents divide labor to achieve a shared goal		
D Independent — agents pursue separate unrelated goals		
40	Collaboration & Environment	A smart city traffic system assigns road clearing routes to emergency vehicles. Multiple route-planning agents bid based on their estimated clearance time. This is:
A Cooperative — all agents share the goal of helping emergencies		
B Competitive — route agents bid against each other for the assignment		
C Adversarial — agents try to block each other's proposed routes		
D Sequential — each agent waits for the previous one to finish		
41	Collaboration & Environment	In a "fully observable" environment for a MAS, what is true?
A Only the orchestrator agent can see all state; sub-agents see partial state		
B Every agent can access all relevant state information at any time		
C The environment is frozen — it does not change while agents deliberate		
D Agents can observe but cannot affect the environment		

42 Collaboration & Environment	When an agent's environment is partially observable, which technique helps the agent make better decisions despite incomplete information?
A	Increasing the LLM temperature to be more creative
B	Adding more sub-agents until full coverage is achieved
C	Using state estimation — inferring hidden state from observable signals via prediction models or Bayesian methods
D	Switching to a competitive collaboration mode
43 Collaboration & Environment	Multiple agents in a manufacturing MAS all read from and write to a shared "defect_log" data store. No agent communicates directly with another — they coordinate through the shared store. This architecture is called:
A	Request-Reply Protocol
B	Contract Net Protocol
C	Blackboard Architecture
D	Pipeline Architecture
44 Collaboration & Environment	A MAS for supply chain disruption must react to real-time geopolitical news, weather events, and supplier outages — the environment changes faster than agents can fully process it. This is best described as:
A	Fully observable and static
B	Partially observable and dynamic
C	Fully observable and dynamic
D	Partially observable and static
45 A2A Protocol & Integration	The Google Agent2Agent (A2A) Protocol is designed to solve which core problem?
A	Training multiple LLM models on the same dataset simultaneously
B	Enabling agents built on different frameworks (LangGraph, CrewAI, ADK, custom) to communicate and collaborate
C	Replacing REST APIs with a faster binary protocol for microservices
D	Encrypting agent communication to prevent data leakage
46 A2A Protocol & Integration	In the A2A Protocol, an "Agent Card" is:
A	A physical ID card that developers must carry to use the A2A API
B	A JSON descriptor that announces an agent's capabilities, endpoint URL, and authentication method
C	A credit card connected to the agent's API billing account
D	A dashboard card that displays agent performance metrics

47 A2A Protocol & Integration	What is the key conceptual difference between MCP (Model Context Protocol) and A2A Protocol?
A	MCP is for Python agents; A2A is for TypeScript agents
B	MCP connects agents to tools, APIs, and resources; A2A enables peer-to-peer communication between agents themselves
C	A2A is deprecated and has been replaced by MCP in all use cases
D	MCP supports real-time streaming; A2A only supports batch processing
48 A2A Protocol & Integration	You have a LangGraph agent that needs to invoke a CrewAI agent as a subcontractor for a specialized task. Which protocol enables this cross-framework delegation?
A	REST API with a custom JSON payload you design yourself
B	Google A2A Protocol — the standard for cross-framework agent communication
C	SOAP web services with WSDL definitions
D	GraphQL subscriptions for real-time agent updates
49 A2A Protocol & Integration	In Google ADK, session.state is used for agents within the SAME application. When should you use A2A instead?
A	When agents are defined in the same Python file
B	When two agents use the same LLM model
C	When agents span different services, frameworks, or vendor systems that do not share a codebase
D	When you want faster communication than session.state allows
50 A2A Protocol & Integration	A production MAS must handle cases where an A2A call to an external agent times out. Which design approach is most appropriate?
A	Crash the entire MAS workflow and require manual restart
B	Ignore the timeout and proceed as if the agent responded successfully
C	Implement fallback logic: retry with exponential backoff, use a cached result, or route to an alternative agent
D	Switch from A2A to FIPA-ACL for the problematic agent connection